

JATIN CHOPRA

Personal Information

Full Name: Jatin Chopra **Nationality:** Indian **Degree Programme:** MSc Astrophysics
Current Institution: Rheinische Friedrich-Wilhelms-Universität Bonn
(Expected) Graduation Date: 15th November 2026
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Education

Rheinische Friedrich-Wilhelms-Universität Bonn Bonn, Germany
MSc Astrophysics October 2023 – November 2026

Savitribai Phule Pune University Pune, India
BSc Physics (Minor in Mathematics and Electronics) July 2019 – September 2022
CGPA: 9.77

Research Internships

- Estimating Initial Cloud Conditions from Molecular Abundances using Neural Networks and Bayesian Inference** *October 2025 – Ongoing*
 - * Supervised by Mr. Gijs Vermariën and Prof. Serena Viti at the Argelander-Institut für Astronomie (AIfA) and Universiteit Leiden. Developing a Neural Network in **JAX** and using **NumPyro** to predict molecular abundances in interstellar gas clouds.
 - * Performing Bayesian Inference to estimate initial cloud conditions, including temperature, final density, cosmic ray ionisation rate, and UV background radiation.
 - * Investigating the effect of incorporating survival function, CDF, and PDF for molecules with only upper-limit abundance measurements, improving uncertainty estimation of molecules with weak observation.
- Studying the Role of Galaxy Spheroids in Star-Formation** *May 2025 – Ongoing*
 - * Supervised by Dr. Dario Colombo and Prof. Frank Bigiel at the Argelander-Institut für Astronomie (AIfA).
 - * Built a Python pipeline for multicomponent photometric decomposition of 643 galaxies (**iEDGE** survey), extracting Bulge-to-Total (B/T) light ratios and photometric bulge masses.
 - * Testing the influence of bulge mass on star-formation quenching, thus testing the significance of morphological quenching.
- Scattering of Methanol Masers and Electron Distribution of the Milky Way** *December 2024 – April 2025*
 - * Research intern under Dr. Andreas Brunthaler and Prof. Amélie Saintonge at Max Planck Institute for Radio Astronomy, Bonn. Reduced and analysed **VLBA** data using **AIPS** and **ParselTongue** to measure angular broadening of 21 CH₃OH masers and 10 background quasars.
 - * Compared measured scattering against the **NE2001** electron density model via dispersion measure calculations. Observed discrepancies between the model predictions and the measured scattering for masers, which are attributed to intrinsic maser sizes. No discrepancies were observed for quasars.
- Detecting and Classifying AGNs in MaNGA Galaxies using Resolved BPT Diagnostics** *July 2024 - September 2024*
 - * Developed a **Python+SQL** pipeline to detect AGNs from the **MaNGA** survey by making resolved BPT maps and classifying AGNs based on H α equivalent width into strong/weak.
 - * Detected 567 AGNs from 10,070 MaNGA galaxies, with good classification overlap with broad, X-ray, and IR AGN catalogues; this project was supervised by Dr. Mallory Thorp and Prof. Frank Bigiel at the Argelander-Institut für Astronomie (AIfA).

Title: Star-Formation Quenching in MaNGA Post-Merger Galaxies

Supervisor(s): Prof. Frank Bigiel and Dr. Mallory Thorp

Submission: 15th November 2026

Thesis Abstract

Context: Galaxy-Galaxy mergers are well-known phenomena that alter the morphology and the star-formation of the constituent galaxies. Many simulation-based studies have shown that the cessation of star-formation (star-formation quenching) is caused by AGN triggered by mergers [1]. However, many other studies have proposed different quenching mechanisms, such as morphological quenching, stellar feedback, and ram-pressure stripping [2, 3]. Yet observations have failed to determine which of these mechanisms dominates galaxy evolution. We aim to investigate the question: Do mergers quench star-formation differently? With integral field spectroscopy, one can investigate the influence of mergers on star-formation. Recent spatially resolved studies have revealed that merger-triggered starbursts have unique star-formation radial profiles compared to their secular counterparts [4], as shown in Figure 1. So what remains unanswered is whether the merger-induced star-formation changes as galaxies transition across different evolutionary stages? Answering this question is essential in understanding the significance of mergers in star-formation quenching.

Aim: We are trying to answer two questions: (1) Are post-mergers distributed differently across quenching stages? (2) Does star-formation quenching in post-mergers differ significantly from non-interacting galaxies?

Methods: We use the integral field spectroscopy data from the Mapping Nearby Galaxies at Apache Point Observatory (MaNGA) Survey (10,070 Galaxies). Using IFU observations of galaxies with a range of star-formation properties, Kalinova et al. 2021 found distinct kpc-scale signatures in a galaxy's ionised gas marking an evolution from star-forming to quenched, which they have labelled with unique "quenching stages" [5]. We classified the galaxies visually into their respective quenching stage, identical to Kalinova; Figure 2 is an illustration of the respective quenching stages. The sample has 388 Post Mergers (PMs) and 7051 Non-Interacting Galaxies (Controls). A Python+SQL-based pipeline is developed for analysing the large dataset. Each post-merger is matched to at least five control galaxies with similar properties at the same quenching stage. Radial profiles of $\Delta\Sigma_{\text{SFR}}$ (Offset from the resolved star-formation main sequence, calculated using $\text{H}\alpha$ emission) are created by comparing star-forming spaxels of PMs and Controls that occupy similar regions within a galaxy, thus ensuring a like-for-like comparison of resolved star-formation activity. Additionally, we identify AGNs with optical diagnostics to probe their influence on the resolved star-formation rates.

Results: The distribution of PMs and Controls across the quenching stages is broadly similar. Notably, PMs show an excess in the Star Forming case (SF) and a deficit in the Fully Retired case (fR) compared to the controls, which are the beginning and the end of the quenching scheme. This suggests that while mergers do not dramatically redistribute galaxies across the quenching stages, they may moderately enhance star-formation and delay retirement. Within each quenching stage, the $\Delta\Sigma_{\text{SFR}}$ radial profiles do not show any significant differences between the PMs and Controls. However, when we examine the profiles of SF galaxies broken into bins of ΔSFR , we reproduce the central offset between mergers and isolated galaxies as seen in Thorp et al. 2024 [4]. This implies that though the magnitude of star-formation enhancement is tied to kpc-scale variations in the ISM related to mergers, star-formation suppression is a more universal process with no merger influence on kpc-scales.

Conclusion: The absence of any significant and unique features in the $\Delta\Sigma_{\text{SFR}}$ radial profiles of PMs and controls leads us to conclude that PMs quench similarly to Non-Interacting galaxies. PMs are believed to be bulge-dominated. However, the excess of SF cases in post-mergers argues against the idea of morphological quenching, which states that the domination of the bulge causes quenching [2]. We are now working with alternative and more continuous measurements of star-formation (not reliant on $\text{H}\alpha$ emission [6, 7]) to create a more complete picture of the resolved quenching scheme of the galaxies and further constrain the significance of mergers in galaxy evolution.

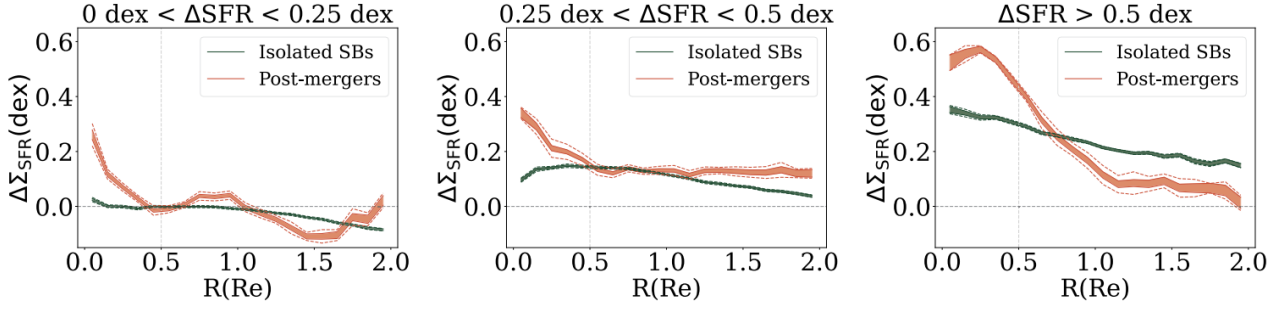


Figure 1: Median radial profiles of $\Delta\Sigma_{\text{SFR}}$ broken into different ΔSFR bins. The starburst post-mergers (PMs) are shown in orange, and the isolated starbursts in green. PMs consistently have a central enhancement in $\Delta\Sigma_{\text{SFR}}$ compared to isolated starbursts, adapted from [4].

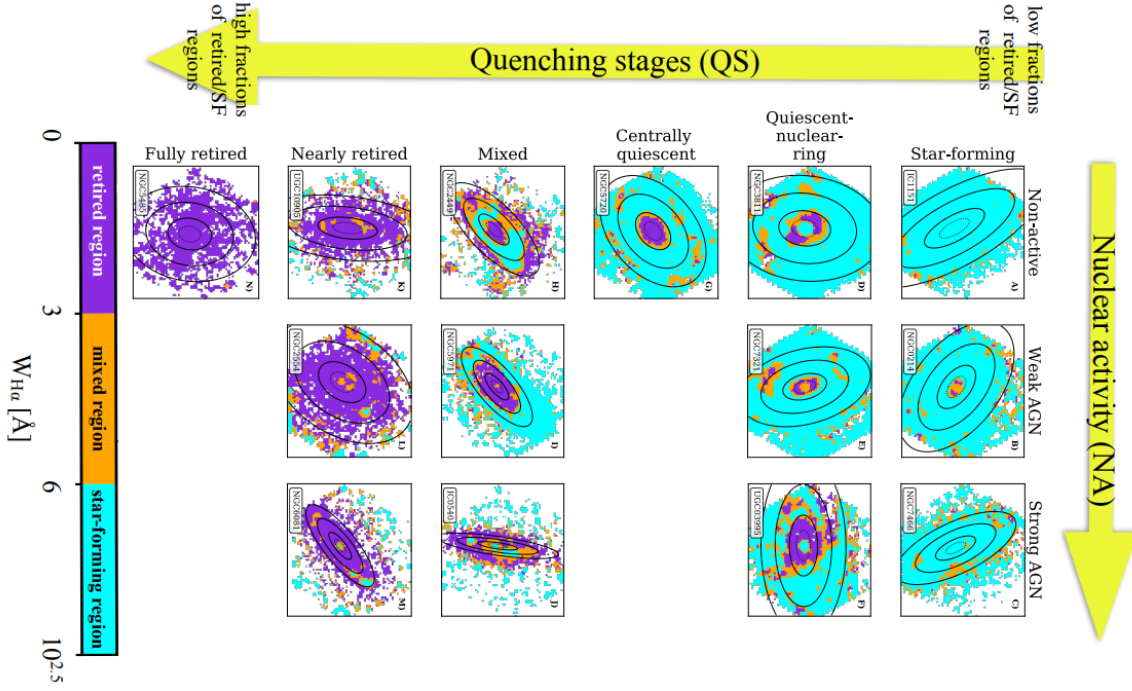


Figure 2: The quenching classification developed using $H\alpha$ equivalent width, adapted from [5]. The classes are (from right to left) Star-forming (SF), Quiescent-nuclear ring (QnR), Centrally quiescent (cQ), Mixed (MX), Nearly retired (nR), and Fully retired (fR). The classes are further classified using nuclear activity (Top to Bottom).

Thesis References

- [1] Philip F Hopkins et al. “A cosmological framework for the co-evolution of quasars, supermassive black holes, and elliptical galaxies. I. Galaxy mergers and quasar activity”. In: *The Astrophysical Journal Supplement Series* 175.2 (2008), pp. 356–389.
- [2] Asa FL Bluck et al. “Bulge mass is king: the dominant role of the bulge in determining the fraction of passive galaxies in the Sloan Digital Sky Survey”. In: *Monthly Notices of the Royal Astronomical Society* 441.1 (2014), pp. 599–629.
- [3] Asa FL Bluck et al. “The quenching of galaxies, bulges, and disks since cosmic noon-A machine learning approach for identifying causality in astronomical data”. In: *Astronomy & Astrophysics* 659 (2022), A160.
- [4] Mallory D Thorp, Sara L Ellison, and Ana Galicia. “Are all starbursts equal? Star-formation-rate profiles in merger versus secular starbursts”. In: *Astronomy & Astrophysics* 690 (2024), p. L4.
- [5] V Kalinova et al. “Star formation quenching stages of active and non-active galaxies”. In: *Astronomy & Astrophysics* 648 (2021), A64.
- [6] Asa FL Bluck et al. “Are galactic star formation and quenching governed by local, global, or environmental phenomena?” In: *Monthly Notices of the Royal Astronomical Society* 492.1 (2020), pp. 96–139.

- [7] Mallory D Thorp et al. “The ALMaQUEST Survey X: what powers merger induced star formation?” In: *Monthly Notices of the Royal Astronomical Society* 516.1 (2022), pp. 1462–1480.

Teaching Experience

- Argelander-Institut für Astronomie (AIfA)** August 2025 - November 2026
Wissenschaftliche Hilfskraft (Tutor of Optical Astronomy Exoplanet Transit Detection Lab), Bonn, Germany
 - * Explaining the observation, calibration, and data reduction of a modern optical telescope, conducting oral examinations, carrying out telescope observation, supervising data reduction, and grading lab reports of MSc students. Additionally, giving tours of the telescope on the rooftop of AIfA to visiting students and guests.
 - * Contributing to the astrophotography done at AIfA by observing Milky Way objects and other deep-sky objects.
- Naxxatra Sciences and Collaborative Research** January 2023 - February 2023
Research and Teaching Fellow, Bangalore, India
 - * Conducted hands-on astrophysics lectures for 300 students (grades 5-12), on topics like galaxy morphology, the cosmic distance ladder, the working of different telescopes, methods to detect exoplanets, and introduced dark matter and CMB.
 - * Performed experiments like sunspot tracking, acceleration due to gravity measurement, and Earth’s circumference calculation. Supervised Solar observations. Conducted stargazing sessions and comet C/2022 E3 (ZTF) observation nights for students, faculty, and parents.

Scientific Training

- Attended the 2nd European ALMA Winter School 2026** January 2026
The school was organised by the University of Leiden, which covered the calibration, imaging, proposal writing, and reduction pipeline to study the astronomical data observed with the ALMA telescope. I was one of the few master’s students to attend this school. I gained an improved understanding of imaging beyond the transitional CLEAN method.
- Attended the Physics of the Early Universe 2022** January 2022
I was one of the very few undergraduate students selected to attend this program organised by the International Centre for Theoretical Sciences (ICTS) Bangalore. Studied advanced topics such as Perturbation Theory, Dark Energy, Inflation, CMB anisotropies, Galaxy Formation, and Big Bang Nucleosynthesis.

Conferences

- Astronomische Gesellschaft (AG) 2026** September 2026
Submitted an abstract of my thesis work for a splinter talk on star-formation and to present a poster at the AG 2026 annual meeting, and am currently awaiting confirmation. View abstract

Honours and Awards

- Selected for the International Summer School on the ISM of Nearby Galaxies** July 2025
I was selected to attend the International Summer School on the ISM of Nearby Galaxies 2025 with partial funding. Unfortunately, I could not attend the school as one of my MSc exams clashed with the school dates.
- Highest Grade in the College Department, BSc Physics** March 2023
Graduated with the top grade in my department at N.W. College of Savitribai Phule Pune University and was awarded a departmental scholarship in recognition of this performance.
- Discovered Two Asteroids** September 2021
I detected near-Earth objects and main-belt asteroids in data from the Pan-STARRS survey, provided by the International Astronomical Search Collaboration (IASC). Made 12 preliminary discoveries and 2 provisional discoveries, including two main belt asteroids, namely, **2021 RZ8** and **2021 QO24**.

Skills

- * **Programming Languages:** Python, SQL, C++, HTML, Fortran
- * **Python Libraries:** NumPy, Pandas, Astropy, Matplotlib, Seaborn, SciPy, pymysql, ParselTongue, JAX, NumPyro, PyTorch, Scikit-learn, sep, photutils, astroquery
- * **Astrophysics Software:** CASA, AIPS, MESA, DS9, PHOEBE, Period04, Astrometrica, NINA
- * **Surveys & Telescopes:** MaNGA, BeSSeL, iEDGE, SDSS, Pan-STARRS, VLBA, ALMA
- * **Other Computational Skills:** Linux, L^AT_EX, GitHub, Microsoft Office
- * **Soft Skills:** Pedagogy and Science Communication, Leadership and Team Collaboration, Analytical Problem Solving, Rapid Skill Acquisition, Resilience and Discipline, Detail-Oriented and Self-Motivated
- * **Languages:** Hindi (Native Speaker), English, German, Punjabi, and Marathi.

Other Relevant Experience

1. **Astrophotography at Universität Bonn** *August 2025 - Ongoing*
Link: Astrophotography Jatin Chopra
Conducted frequent deep sky observations using the 35 cm optical telescope at AIfA. Observed objects like NGC 891, M51, M81, M82, M1, and NGC 7331.
2. **Writer at The Universe 101** *December 2020 - April 2021*
Website: The Universe 101
Began an astrophysics blog to explain cosmic mysteries in simple terms. Articles like "Cosmic Alchemy" and "Dark Matter" received decent attention. I plan to restart the blog and eventually transform it into a YouTube channel.

References

- * **Prof. Frank Bigiel**
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- * **Prof. Serena Viti**
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- * **Dr. Mallory Thorp**
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